## Learning Objectives

Upon completion of this chapter, students will be able to

- apply scalar and vector products of two and three vectors
- solve problems in geometry, trigonometry, and physics
- derive equations of a line in parametric, non-parametric, and cartesian forms in different situations
- derive equations of a plane in parametric, non-parametric, and cartesian forms in different situations
- find angle between the lines and distance between skew lines
- find the coordinates of the image of a point

## 6.2 Geometric introduction to vectors

A vector $\vec{v}$ is represented as a directed straight line segment in a 3-dimensional space $\mathbb{R}^3$, with an initial point $A = (a_1, a_2, a_3) \in \mathbb{R}^3$ and an end point $B = (b_1, b_2, b_3) \in \mathbb{R}^3$, and it is denoted by $\overrightarrow{AB}$. The length of the line segment $AB$ is the magnitude of the vector $\vec{v}$ and the direction from $A$ to $B$ is the direction of the vector $\vec{v}$. Hereafter, a vector will be interchangeably denoted by $\vec{v}$ or $\overrightarrow{AB}$. Two vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ in $\mathbb{R}^3$ are said to be equal if and only if the length $AB$ is equal to the length $CD$ and the direction from $A$ to $B$ is parallel to the direction from $C$ to $D$. If $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are equal, we write $\overrightarrow{AB} = \overrightarrow{CD}$, and $\overrightarrow{CD}$ is called a **translate** of $\overrightarrow{AB}$.

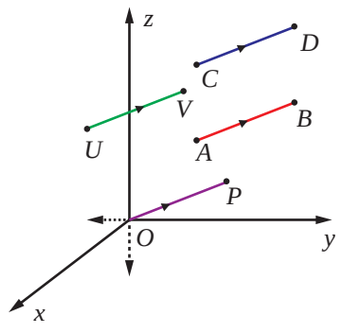


Fig. 6.1

It is easy to observe that every vector $\overrightarrow{AB}$ can be translated to anywhere in $\mathbb{R}^3$, equal to a vector with initial point $U \in \mathbb{R}^3$ and end point $V \in \mathbb{R}^3$ such that $\overrightarrow{AB} = \overrightarrow{UV}$. In particular, if $O$ is the origin of $\mathbb{R}^3$, then a point $P \in \mathbb{R}^3$ can be found such that $\overrightarrow{AB} = \overrightarrow{OP}$. The vector $\overrightarrow{OP}$ is called the **position vector** of the point $P$. Moreover, we observe that given any vector $\vec{v}$, there exists a unique point $P \in \mathbb{R}^3$ such that the position vector $\overrightarrow{OP}$ of $P$ is equal to $\vec{v}$. A vector $\overrightarrow{AB}$ is said to be the **zero vector** if the initial point $A$ is the same as the end point $B$. We use the standard notations $\hat{i}, \hat{j}, \hat{k}$ and $\vec{0}$ to denote the position vectors of the points $(1,0,0), (0,1,0), (0,0,1)$, and $(0,0,0)$, respectively. For a given point $(a_1, a_2, a_3) \in \mathbb{R}^3$, $a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ is called the position vector of the point $(a_1, a_2, a_3)$, which is the directed straight line segment with initial point $(0,0,0)$ and end point $(a_1, a_2, a_3)$. All real numbers are called scalars.



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